

## SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

**Authority.** Corrected French TeX lines 3429–3431; original printed p. 116, physical source-PDF p. 101, and recomposed running p. 93. These locator systems are kept distinct. Blank line 3432 is excluded; the raw cursor is line 3432 and the next substantive cursor is line 3433 (the proof). The complete corollary, covering type, fiber product, connectivity equivalences, and symbols were checked directly. The external English candidate is comparison-only.

**Corollary 2.4.** Suppose that  $\text{Lef}(X, Y)$  holds. A finite flat covering  $R$  of an open neighborhood  $U$  of  $Y$  is connected if and only if  $R \times_U \widehat{X}$  is connected. In particular,  $Y$  is connected if and only if  $U$  is connected, and this is also equivalent to  $X$  being connected.